An excitatory and inhibitory neural field model can be written as:

$$\begin{aligned}\tau_e u_t(x,t) &= -u + F(W_{ee}(x) * u - W_{ie}(x) * v + I_e(x,t) - \theta_e) \\ \tau_i v_t(x,t) &= -v + F(W_{ei}(x) * u - W_{ii}(x) * v + I_i(x,t) - \theta_i)\end{aligned}$$

where $\tau_j$ are time constants, $u$ is the excitatory and $v$ is the inhibitory population. $W(x) * u$ means the convolution of $W(x)$ with $u(x,t)$ where $x$ is some spatial location in one or two dimensions.

1. In some old work (2011) Rule et al looked at uniform flicker inputs $I(x,t) = H(\cos(2\pi t/P) - c)$ where $H$ is the step function. They showed that with certain parameter choices, it is possible to evoke spatial hallucinations at due to the flicker. With some undergraduate students, this summer, we extended this model to incorporate a simple model for color opponency based on a recent paper by Faugueros et al. We found some nice examples of color hallucinations. There is much analysis that can be done on some simplified versions of this and also on the full model. Division of the stimulus parameters into different regimes of bifurcation has not yet been done. I have some python/matlab/xpp codes for the spatial model. Another aspect is to see how different colors of light might also affect the patterns.

2. Arnold Wilkins and others have reported extensively on uncomfortable images; these are images whose properties (such as high contrast stripes) are not easy to look at. In some people, these types of patterns cause "pattern sensitive epilepsy" (PSE). A PhD student did some extensive work on this about 10 years ago (and some appeared in his thesis), but there is much left to do and I'd like to get something on this done before I retire. In this case, $I(x,t)$ consists of spatially periodic patterns but constant in time. We found that some spatial patterns become unstable and lead to complex oscillations. Some theory has been done but more is needed and some of the old simulations appear to be incorrect.

3. These two problems are examples of the interaction of external patterns with a spatio-temporal pattern forming system. For a more analytically inspired student, I'd like to propose a calculation. I will write this generally, but the above system can be written in this form. Consider:

   $$u_t = L_0 u + \lambda L_1 u + Q(u,u) + C(u,u,u) + \epsilon^n P e^{i\omega t} + \epsilon^m Q e^{ikx} + c.c.$$

   $P, Q$ are complex vectors and $n, m, k$ are natural numbers. $Q, C$ are quadratic and cubic forms. For simplicity, assume a 1-d periodic spatial domain on $[0, 2\pi]$. The operator $u_t - L_0 u$ has a four-d nullspace spanned by $\Phi \exp(i[\omega_0 t \pm k_0 x])$ plus complex conjugates. Without the forcing terms, one can write down the normal form equations to see which patterns are selected (e.g. standing vs traveling waves) (See Curtu & Ermentrout, 2004, SIADS for a neural specific implementation). The question is how do the forcing terms interact with this pattern formation? What values of $n, m$ are needed and how do they depend on $\omega, k$ relative to $\omega_0, k_0$.